MATTHEW EYRE

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EDUCATION

University of Waterloo, Waterloo, ON Sep 2025 - Apr 2030
Bachelor of Applied Science, Mechanical Engineering | GPA: 88.54%

TECHNICAL SKILLS

CAD & Drafting: SOLIDWORKS (3D modeling, assemblies, 2D engineering drawings, Simulation), AutoCAD, GD&T, tolerancing, bills of materials
Manufacturing & Prototyping: Machining (lathe, mill, drill press, bandsaw), 3D printing, laser cutting, assembly, prototyping, machine shop certified
Testing & Integration: Experimental test setups, calibration, vibration testing, materials characterization, motor and sensor integration, debugging, MATLAB, Excel
Programming & Tools: C++, MATLAB, Python, CMiC, technical documentation

TECHNICAL EXPERIENCE

Govan Brown Building Group, Toronto, ON May 2026 - Aug 2026
Project Administrator, Co-op
- Set up and tracked the **MEP testing, adjusting, and balancing (TAB)** plan; conducted site visits with trades to verify testing and balancing was performed properly, and reviewed TAB commissioning documents
- Reviewed architectural, mechanical, and electrical **drawings** to verify construction compliance; documented deficiencies using photography, drawings, and markups during site inspections
- Developed suite-level tracking system to organize **1,500+ deficiencies**; independently managed tracking for entire 5-story building across **42 suites and 37 trades**

UW Orbital Design Team, Waterloo, ON Sep 2025 - Present
Mechanical Member
- Built **CubeSat structural frame** and individual components in **SOLIDWORKS**; created and organized the complete mechanical **assembly** and component interfaces, including internal threading
- Translated system-level requirements into a manufacturable assembly, considering fastener placement, assembly accessibility, and **dimensional constraints**
- Perform **vibration testing** to assess structural and fastener integrity; collaborate with multidisciplinary team on component development and testing

PROJECTS

Stash&Store Automated Coin Management System (ME 101) Sep 2025 - Apr 2026
- Designed, manufactured, and programmed a fully functional **ATM**: mechanical casing, coin sorting/dispensing mechanisms, electronic mounts for motors and optical sensors, and user interface; built using laser cutting, 3D printing, and manual fabrication
- Designed **kinematic coin-pushing mechanisms** (crank shaft, piston arm, coin pusher) around **mechanical tolerances** and iteratively tested prototypes to eliminate jams and reliably handle different coin sizes; achieved **98% dispensing accuracy**
- Integrated VEX motors, **optical and inertial sensing** in **C++** with modular software; earned 100% on final project

Flywheel-Driven Launch System and Motor Dynamometer Sep 2025 - Apr 2026
- Designed and prototyped flywheel-driven mechanical launch system in CAD with **energy-transfer and safety-factor analysis**, considering motion, assembly, and manufacturability
- Designed dynamometer **test setup** to characterize **motor torque-speed performance**; developed test procedures, **calibrated components**, and collected experimental data

Machined Keychain Assembly
- Designed and machined a multi-part keychain assembly (aluminum plates and spacer, acrylic spur gear); produced **SOLIDWORKS** assembly and 2D **engineering drawing with bill of materials**

Materials Testing Laboratory Work
- Ran tensile tests on steel, aluminum, and copper and cantilever beam deflection tests to determine **Young's modulus, yield strength, and ultimate tensile strength**; characterized **creep of Sn-Zn solder** at 22 and 50 deg C (stress exponent n = 5.38)

LEADERSHIP & EXTRACURRICULARS

Action CIND Support Group, Co-Founder: Co-founded and facilitate bi-weekly online support sessions for 10-15 participants; develop discussion topics, resources, and activities
Diabuddies, Hospital Volunteer: Support newly diagnosed children with Type 1 Diabetes and their families at volunteer events
Habitat for Humanity, Volunteer: Volunteered to help build homes for families in need (July 2026)

AWARDS & CERTIFICATIONS

Awards: President's Scholarship of Distinction, Engineering Entrance Scholarship, Tamera Gordon Foundation Award
Certifications: WHMIS, Engineering Student Machine Shop Training